Algebra 1 Name Answer Key____________
Unit 10 Date___________________
Lesson H2 Homework Period__________________

Use the following context to answer questions 1-4.

Members of the cross-country team are preparing for a 3-mile race. The average time for the team is 20.5 minutes. The fastest and slowest times varied from the average by 3.4 minutes.

1. Define the variable.

   Let m represent the minutes ran.

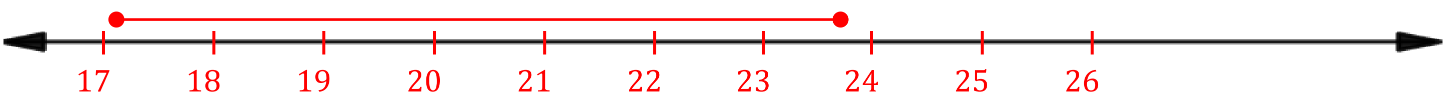
2. Write an absolute value inequality that represents the context.

   $|m - 20.5| \leq 3.4$

3. Solve the inequality. Write the answer as a compound inequality.

   $m - 20.5 \leq 3.4$ $\quad$ $m - 20.5 \geq -3.4$
   $+20.5 \quad +20.5$ $\quad$ $+20.5 \quad +20.5$
   $m \leq 23.9$ $\quad$ $m \geq 17.1$ $\quad$ $17.1 \leq m \leq 23.9$

4. Graph the solution on the number line.



Use the following context to answer questions 5-6.

Franklin is a member of the high school band. Each day he spends an average of 55 minutes practicing for an upcoming band concert. He spent less time practicing on school days, but more time on the weekends. Each of his daily practice times was within 27 minutes of his average.

5. Write an absolute value inequality that represents m, the number of minutes Franklin spends practicing for the band concert on a given day.

   $|m - 55| \leq 27$

6. Solve the inequality.

   $m - 55 \leq 27$ $\quad$ $m - 55 \geq -27$
   $+55 \quad +55$ $\quad$ $+55 \quad +55$

   $m \leq 82$ $\quad$ $m \geq 28$

   $28 \leq m \leq 82$

Use the following context to answer questions 7-10.

Hailey made a personal goal of earning $\$130$ per week at her after-school job. Each week, she has been averaging within $\$23.15$ of her goal.

7. Define the variable.

   Let x represent the money Hailey made.

8. Write an absolute value inequality that represents the context.

   $|x - 130| \leq 23.15$

9. Solve the inequality. Write the answer using set-builder notation.

   $-23.15 \leq x - 130 \leq 23.15$
   $106.85 \leq x \leq 153.15$

   $\{x | 106.85 \leq x \leq 153.15\}$

10. Graph the solution on the number line.